

RVs

① Bernoulli Trials

$$P[\text{success}] = p$$

↙ Case A: fixed number of repetitions
Binomial $X = \text{number of successes}$

$$P[X=x] = \binom{n}{x} p^x (1-p)^{n-x}$$

$$F[X=x] = \sum_{i=0}^x \binom{n}{i} p^i (1-p)^{n-i}$$

$$E[X] = \sum_{i=0}^n i P_X(i)$$

↙ Case B: repeat exp until success
Geometric $X = \text{no. of tries for success.}$

$$P[X=x] = p(1-p)^{x-1}$$

$$F[X=x] = \sum_{i=1}^x p(1-p)^{i-1}$$

$$E[X] = 1/p \quad , \quad \text{Var}[X] = \frac{1-p}{p^2}$$

case C: repeated until a fixed no. of success
 Negative Binomial / Pascal
 r = fixed no. of req. successes
 Y = number of exp.

$$P[Y=y] = \binom{y-1}{r-1} p^r (1-p)^{y-r}$$

$$P[Y=y] = \binom{y}{r} p^r (1-p)^{y-r} - \binom{y-1}{r} p^r (1-p)^{y-r}$$

$$\left(\frac{1-p}{1-p} - 1 \right) \frac{\binom{y}{r} p^r (1-p)^{y-r}}{\binom{y-1}{r} p^r (1-p)^{y-r}} = [X] \text{ not}$$

② Hypergeometric Distribution

in a box of r red balls and g green balls, pick n balls at random without replacement

X = no. of red balls picked.

$$P(X=x) = \frac{\binom{n}{x} \binom{g}{n-x}}{\binom{r+g}{n}}$$

$$E[X] = \frac{nr}{r+g}$$

$$\text{Var}[X] = \frac{n(rg)}{(r+g)^2} \left(1 - \frac{n-1}{r+g-1}\right)$$

③ Poisson RV

$$P[X=x] = \begin{cases} \frac{(\lambda T)^x e^{-\lambda T}}{x!}, & x \geq 0 \\ 0, & \text{o/t} \end{cases}$$

λ = rate of occurrence in a time period

T = length of time period

$\therefore \lambda T$ = no. of occurrences.

$$E[X] = \lambda T$$

$$\text{Var}[X] = \lambda T$$

$$\frac{\lambda T}{1} = E[X]$$

$$\frac{(\lambda T + 1 - 1)(\lambda T + 1 - 1)}{1} = \lambda T = \text{Var}[X]$$

④ Uniform RV

$$P_x(x) = \begin{cases} \frac{1}{m-l+1}, & \text{for } x = l, l+1, l+2, \dots \\ 0, & \text{otherwise} \end{cases}, m$$

$$F_x(x) = \begin{cases} 0, & x < l \\ \frac{x-l+1}{m-l+1}, & x = l, l+1, \dots, m-1 \\ 1, & x \geq m \end{cases}$$

$$E[X] = \frac{m+l}{2}$$

$$\text{Var}[X] = \frac{(m-l)(m-l+1)}{12}$$